What I built, in three sentences

1

Catch problems first

Interruption data arrives messy. I built validation rules that catch problems before they reach a reliability or GSL calculation.

2

Rebuild the threshold

I rebuilt Powercor's Major Event Day threshold from its published daily data and matched all 16 MED days.

3

Reconcile the figures

I reconciled Powercor's AER model against the ESC's published GSL figures: 10 of 10 comparable rows match to the dollar.

Real data, four sources

Source	What it holds	Used for
NZ Commerce Commission	37,538 interruption events, 29 distributors, 3,498 feeders	Validation rules, cause mapping
Powercor via the AER	1,827 days of SAIDI, SAIFI and MAIFI, 2019 to 2024, with MED flags	MED threshold, STPIS targets
Powercor GSL model 9.04	GSL counts, rates and dollars by category, FY22 to FY24	GSL reconciliation
ESC Victoria	Published GSL counts and dollars by distributor	Reconciliation counterpart

Stack: SQL Server with staging, clean and reporting layers, Power Query, and a Power BI star schema with DAX measures.

Data quality: 870 exceptions caught before any calculation

870

exceptions logged by 6 rules, across 868 of 37,538 events

617

exact duplicate rows, all from Westpower

369.1

Westpower SAIDI with the duplicates, 184.6 once removed

By rule: exact duplicates 617, momentary events in sustained data 102, extreme duration 75, fractional customer counts 35, missing or zero customers 31, end before start 10. The duplicates are in the published source, not in the load.

A trap in the customer minutes check

7

distributors report customer minutes per customer, not per ICP as the data dictionary says

12,900

alerts a naive 5% mismatch rule would raise, about 99% of them false

89

events report more than is physically possible: the real errors

Most mismatches have a legitimate explanation: per customer reporting, or supply restored to customers in stages. Rules V06 and V02 are designed in the rules catalogue but not applied yet.

Major Event Days: threshold rebuilt, 16 of 16 days match

6.759

minutes: the MED threshold, rebuilt from Powercor's own daily data

Step	Result
Days analysed	1,827
Alpha, mean of ln(SAIDI)	-2.0113
Beta, standard deviation	1.4008
Threshold: e^(alpha + 2.8 × beta)	6.759 min
MED days matching Powercor	16 of 16

31 January 2020: raw SAIDI of 6.984 is above the threshold, but 3.950 after exclusions, so it is not a MED. Exclusions are applied first, then the day is tested.

STPIS targets reproduced exactly

Feeder class	Target: 5 year average, MED days removed	Average if MED days are kept
Urban	56.378	72.8
Short Rural	94.273	150.9
Long Rural	208.154	376.5

Storm days drive rural reliability

Long Rural FY2022 was 481.8 minutes with MED days and 197.4 without: about 59% of rural outage minutes fell on storm days.

GSL reconciliation: Powercor’s model against the ESC

10 of 10 comparable category rows match to the dollar, FY22 and FY23 actuals

Scope difference

Street lights appear only in Powercor’s model ($750 FY22, $575 FY23). The ESC table does not publish them.

Internal check

Rate × count = amount on every actual row, with 0 exceptions.

Label trap

Powercor’s “Low reliability” payments are the ESC’s “Supply restoration”. They were mapped by meaning, not wording.

Storm link

MED day GSL count was 27,479 in FY22 against 7,579 in FY23, matching 5 MED days against 2.

Limitations and next steps

Limitations

- The outage events are New Zealand data, not Powercor’s own.
- Rules V02 and V06 are designed but not applied.
- The disclosure year rule is not built.
- Customer level data is not modelled.
- ESC and Powercor figures are likely not independent.

Next steps

- GSL eligibility engine on modelled customers
- Reconciling indices per distributor
- Publishing the SQL and documentation